

[Edit](#)

Perform Natural Language Processing by Using Amazon Comprehend

Challenge Overview

Understand the scenario

You are a cloud Administrator for Hexelo, an organization that needs to implement auditing for their AWS environment. You have decided to implement the AWS Audit Manager. First, you will configure sample resources for auditing. Next, you will configure the AWS Audit Manager, and then you will create an Assessment. Finally, you will generate an assessment report.

Navigating the Challenge Lab

🔗 ► Understanding a Cloud Slice

🔗 ► Quick tips for navigating the Challenge Lab instructions.

[New to Challenge Labs? Click here to learn more.](#)

Set Up Your AWS Account and IAM Role

Hints Enabled

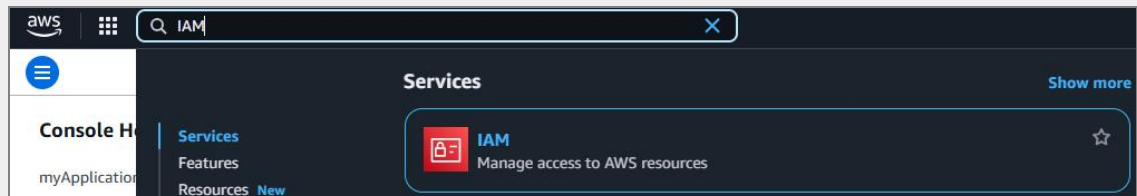
No Yes

- Create an IAM user named T **User01** by using the values in the following table. For any property that is not specified, use the default value.

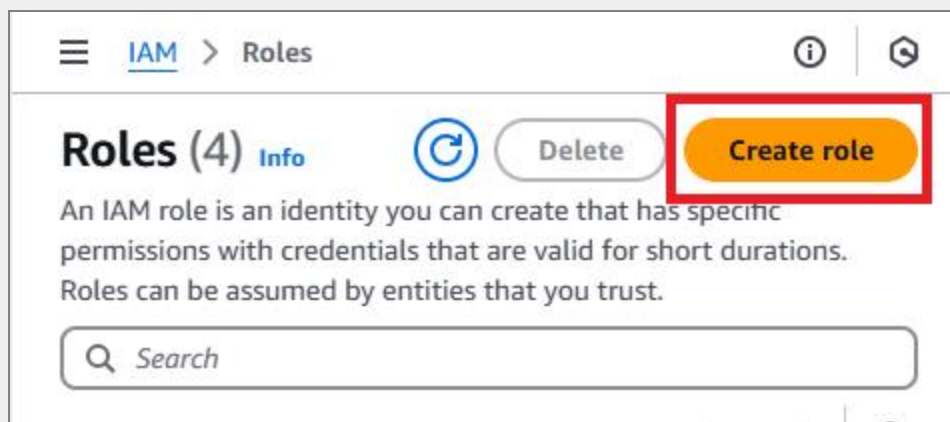
Property	Value
Service	T IAM
Trusted entity type	AWS service
Service type	Amazon EC2
Permissions policy	T ComprehendFullAccess
Role name	T User01

💡 Expand this hint for guidance on creating an IAM user.

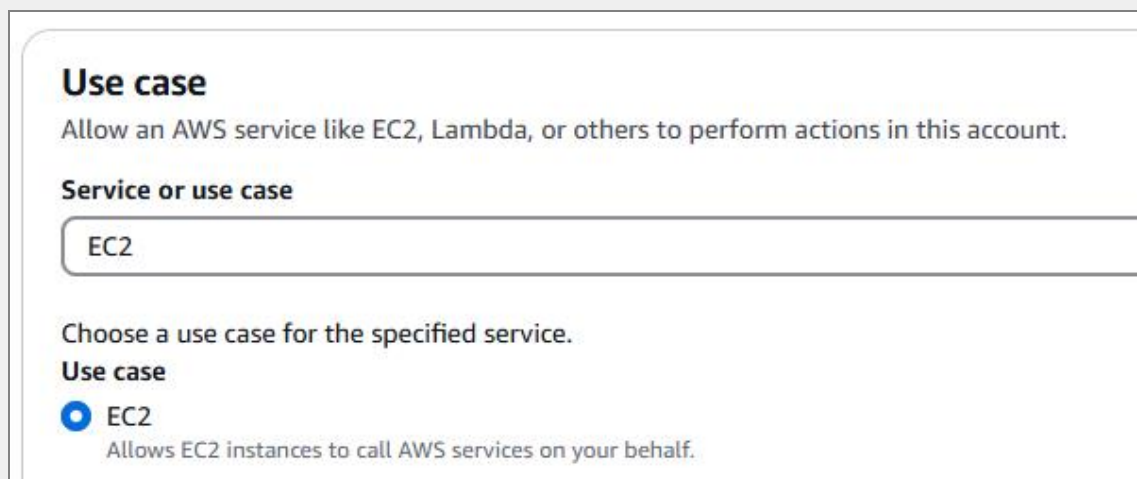
- If necessary, log in to the [AWS Management Console](#).
- In the Services menu, search for and select **IAM**.



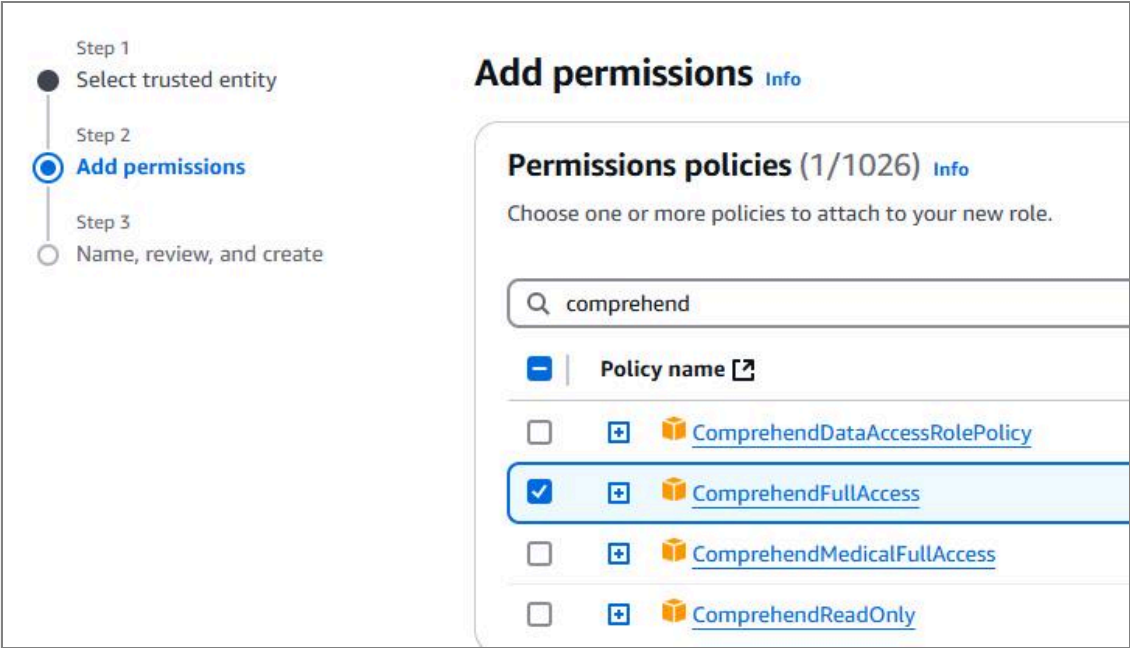
- In the IAM dashboard, select **Roles**, and then select **Create Role**.



- On the Create role page, in Select trusted entity, in Trusted entity type, select **AWS service**.
- In **Use case > Service or use case**, in the *Choose a service or use case* dropdown, select **Amazon EC2**, and then select **Next**.

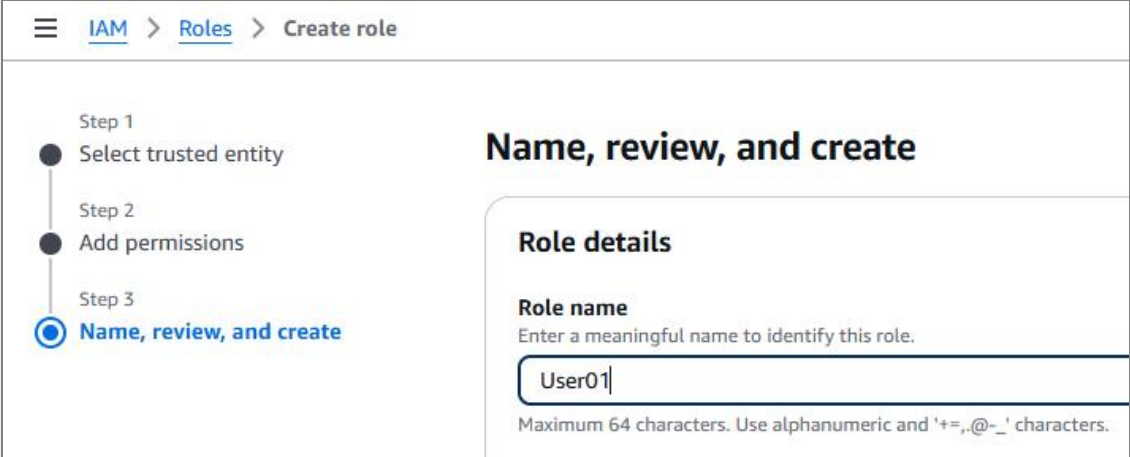


- On the Add permissions page, in Permissions policies, search for and then select the **ComprehendFullAccess** policy, and then select **Next**.



The screenshot shows the 'Add permissions' step in the AWS IAM console. On the left, a progress bar indicates three steps: 'Step 1: Select trusted entity', 'Step 2: Add permissions' (which is the current step), and 'Step 3: Name, review, and create'. The main area is titled 'Add permissions' with an 'Info' link. Below this, it says 'Permissions policies (1/1026)' with another 'Info' link. A search bar contains the text 'comprehend'. A list of policies is shown, each with a checkbox, a plus icon, and a policy name. The policy 'ComprehendFullAccess' is selected with a checked checkbox. Other policies listed are 'ComprehendDataAccessRolePolicy', 'ComprehendMedicalFullAccess', and 'ComprehendReadOnly'.

- On the Name, review, and create page, in Role details, in Role name, enter **T** User01, and then select **Create Role**.



The screenshot shows the 'Name, review, and create' step in the AWS IAM console. The breadcrumb navigation at the top reads 'IAM > Roles > Create role'. The progress bar on the left shows three steps: 'Step 1: Select trusted entity', 'Step 2: Add permissions', and 'Step 3: Name, review, and create' (which is the current step). The main area is titled 'Name, review, and create'. Under the 'Role details' section, there is a 'Role name' field with the text 'User01' entered. Below the field, a note states: 'Maximum 64 characters. Use alphanumeric and '+,=, @, _' characters.'

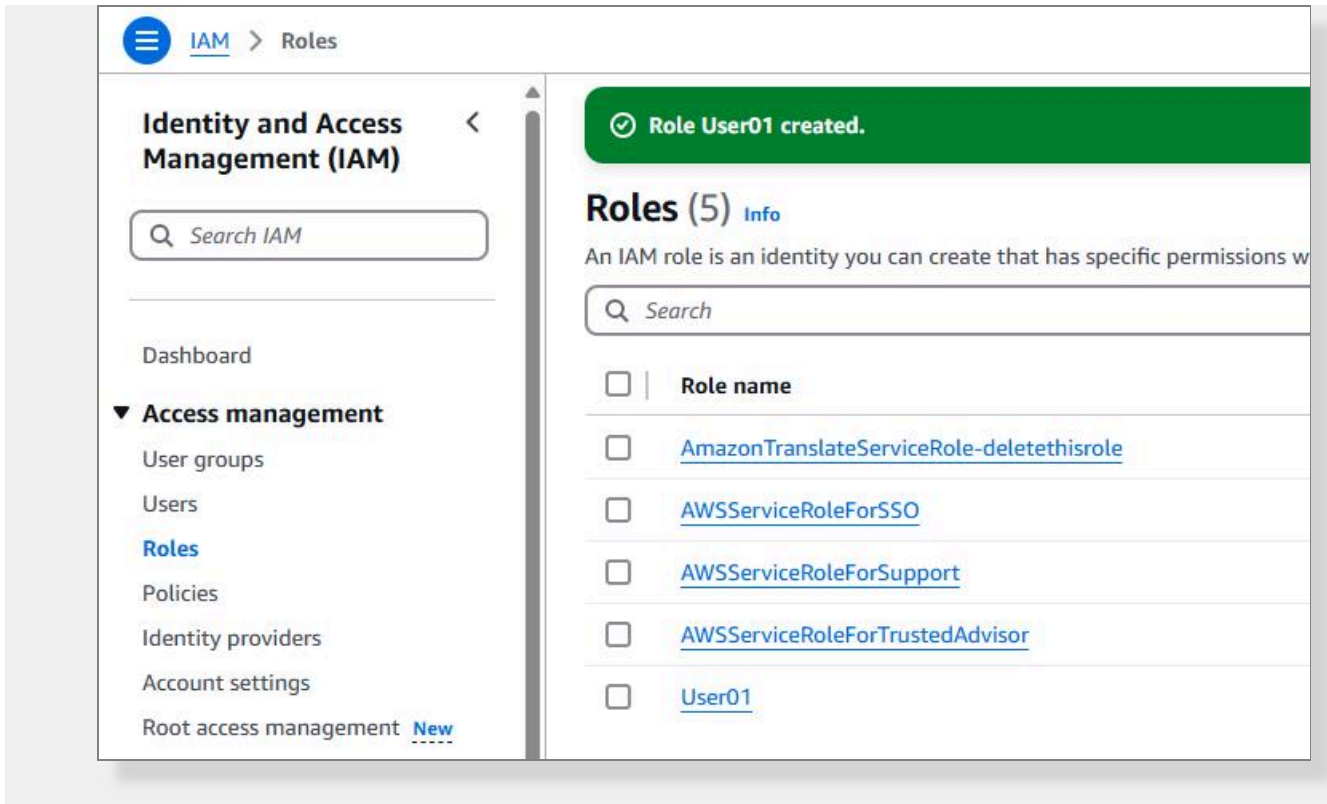


Want to learn more? Review the documentation on [creating an IAM user](#).



Your screen should resemble the following:





- Create a new folder named **T Lesson01** in the *C:* folder.

 Expand this hint for guidance on creating a new folder. 

- Open File Explorer, and then in **This PC**, navigate to the **C:** folder, right-click, select **New**, and then select **Folder**.
- Name the folder named **T Lesson01**, and then select **Enter**.

- Configure the AWS CLI MSI installer for Windows.

💡 Expand this hint for guidance on configuring AWS CLI.

- Navigate to [AWS CLI](#), and then on the AWS Command Line Interface home page, select **Get started**.

AWS Command Line Interface

The AWS Command Line Interface (AWS CLI) is a unified tool to manage your AWS services. With just one tool to download and configure, you can control multiple AWS services from the command line and automate them through scripts.

[Get started](#)[Command Reference](#)

- On the Getting started with the AWS CLI page, in the navigation menu to the left, in Get started, select **Install/Update**.

AWS Command Line Interface

User Guide for Version 2

► About the AWS CLI

▼ Get started

Prerequisites

Install/Update

Past releases

Build and install from source

Amazon ECR Public/Docker

Setup

- On the Installing or updating to the latest version of the AWS CLI page, in AWS CLI install and update instructions, select **Windows**, and then proceed to download the installer to the **C:\Lesson01** folder and run the [AWS CLI MSI installer for Windows \(64-bit\)](#)

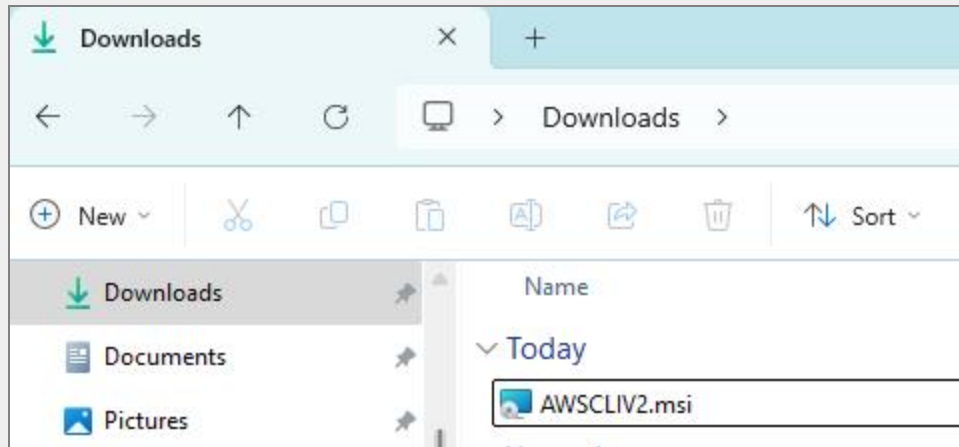
Install or update the AWS CLI

To update your current installation of AWS CLI on Windows, download a new installer each time you update to overwrite previous versions. AWS CLI is updated regularly. To see when the latest version was released, see the [AWS CLI version 2 Changelog](#) on [GitHub](#).

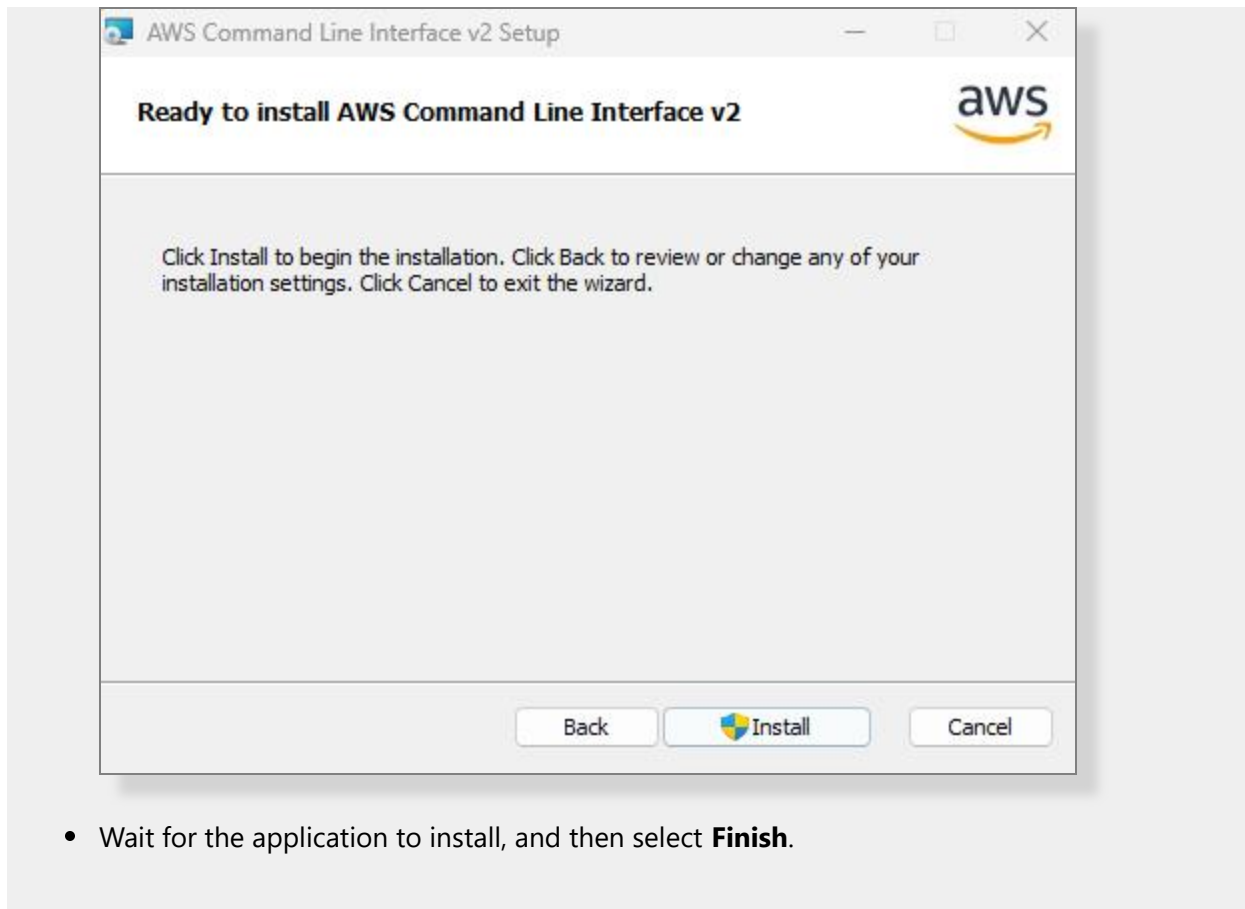
1. Download and run the AWS CLI MSI installer for Windows (64-bit):

<https://awscli.amazonaws.com/AWSCLIV2.msi>

- Navigate to the **C:\Lesson01** folder, and then select the **AWSCLIV2.msi** file to run it.



- In the AWS Command Line Interface v2 Setup page, in Welcome to the AWS Command Line Interface v2 Setup Wizard, select **Next**.
- On the End-User License Agreement page, select **I accept the terms in the License Agreement**, and then select **Next**.
- On the Custom Setup page, review the installation location, and then select **Next**.
- On the Ready to install AWS Command Line Interface v2 page, select **Install**.



 Want to learn more? Review the documentation on [configuring AWS CLI](#).

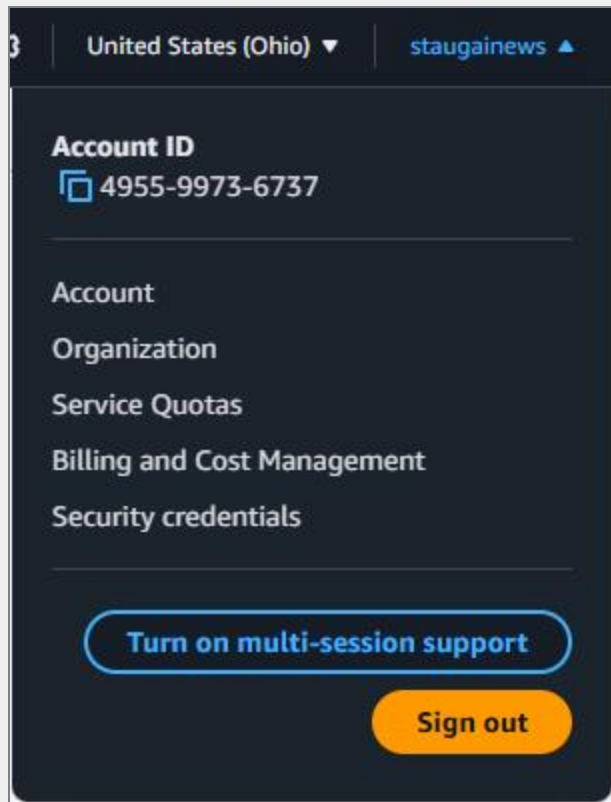
 You can also run the `msiexec` command to run the MSI installer. 

 `C:\> msiexec.exe /i https://awscli.amazonaws.com/AWSCLIV2.msi`

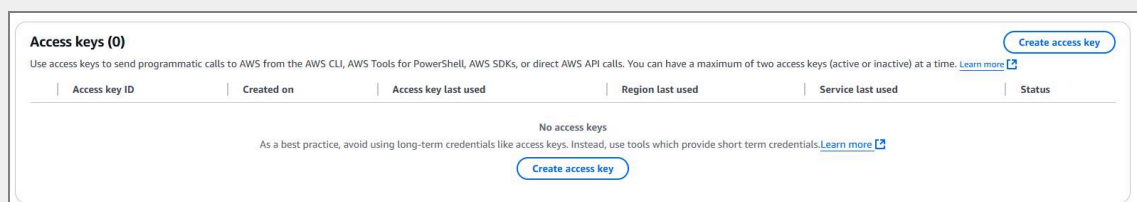
- Create a new Access key.

💡 Expand this hint for guidance on creating an Access key.

- On the AWS Console Home page, in the upper right-hand corner, select your Username, and then select **Security credentials**.



- On the Security credentials page, in My security credentials, scroll down, and then in Access keys, select **Create access key**.



- On the Create access key page, in Alternatives to root user access keys, in Continue to create access key, select **I understand creating a root access key is not a best practice, but I still want to use one**, and then select **Create access key**.

Alternatives to root user access keys [Info](#)**Root user access keys are not recommended**

We don't recommend that you create root user access keys. Because you can't specify the root user in a permissions policy, you can't limit its permissions, which is a best practice.

Instead, use alternatives such as an IAM role or a user in IAM Identity Center, which provide temporary rather than long-term credentials. [Learn More](#)

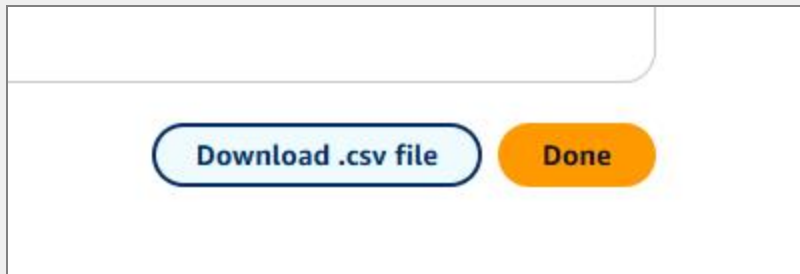
If your use case requires an access key, create an IAM user with an access key and apply least privilege permissions for that user. [Learn More](#)

Continue to create access key?

☒ I understand creating a root access key is not a best practice, but I still want to create one.

[Cancel](#)[Create access key](#)

- Select **Download .csv file**, and then select **Done**.

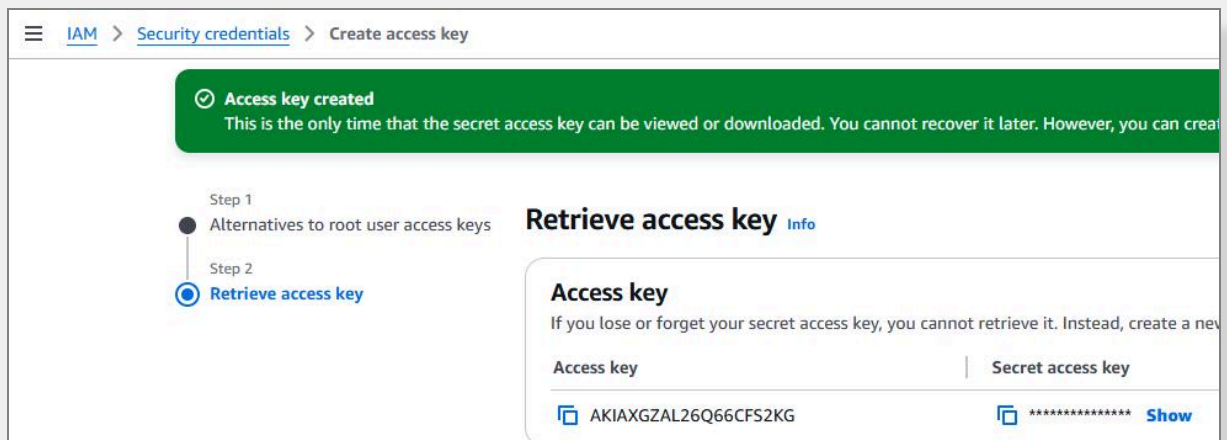


- If necessary, move **rootkey.csv** to the **C:\Lesson01** folder.



Want to learn more? Review the documentation on [creating an Access key](#).

Your screen should look like the following:



- Record the Access key in the following text box.

Access key

 Expand this hint for guidance on locating the Access key. 

- In **rootkey.csv**, in the Access key column, in cell **A2**, copy the Access key, and then paste it into the Access key box above.

- Record the Secret access key in the following text box.

Secret access key

 Expand this hint for guidance on locating the Secret access key. 

- In **rootkey.csv**, in the Secret access key column, in cell **B2**, copy the Secret access key, and then paste it into the Secret access key box above.

- Run  `aws configure` in Powershell, as an administrator, by using  `us-east-1` as the region, and then enter your **AWS access key ID**, **Secret access key**, and assign  `text` as the **Output format**.

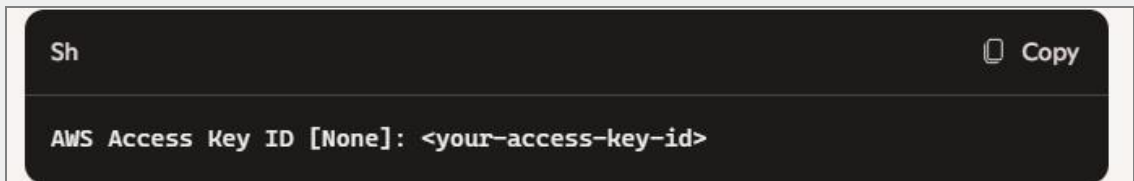
💡 Expand this hint for guidance on running aws configure in your terminal. ^

- Launch **Powershell** as an Administrator, and then in the console, run the following command:

```
powershell
```

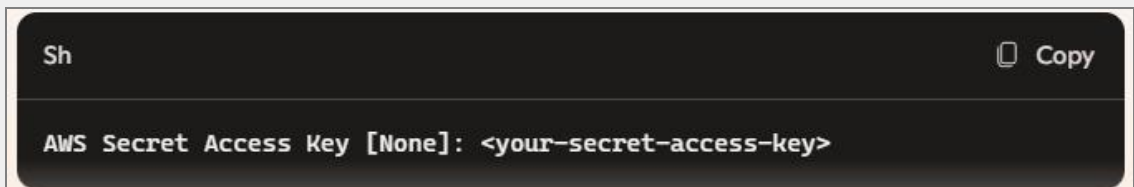
 `aws configure`

- When prompted to enter your AWS Access Key ID, navigate to the Downloads folder, and then open the **rootkey.csv** file.
- In rootkey.csv, in the Access key ID column, select and copy the value in cell **A2**, and then in the Powershell console, right-click to paste the AWS Access Key ID.



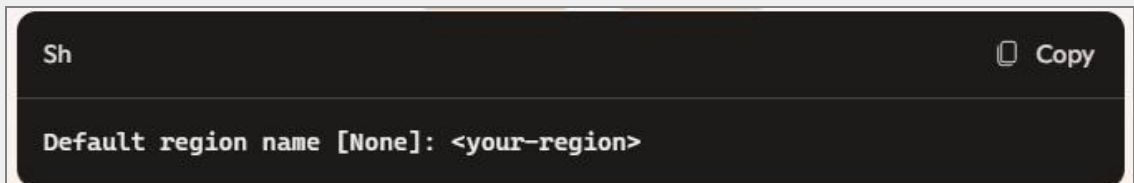
```
Sh AWS Access Key ID [None]: <your-access-key-id>
```

- Select **Enter**.
- In rootkey.csv, in the Secret Access key column, select and copy the value in cell **B2**, and then in the Powershell console, right-click to paste the AWS Secret Access Key.



```
Sh AWS Secret Access Key [None]: <your-secret-access-key>
```

- In Default output format, enter **T** us-east-1, and then select **Enter**.



```
Sh Default region name [None]: <your-region>
```

- In Default output format, enter **T** text, and then select **Enter**.

```
PS C:\Users\DamianKemper> aws configure
AWS Access Key ID [None]: AKIAXGZAL26Q66CFS2KG
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]:
PS C:\Users\DamianKemper>
```



Want to learn more? Review the documentation on [running aws configure in your terminal](#).



Alternatively, you can run the `msiexec` command to run the MSI installer.



```
C:\> msiexec.exe /i https://awscli.amazonaws.com/AWSCLIV2.msi
```

Check your work

Verify

Perform Sentiment Analysis

Hints Enabled

No Yes

Amazon Comprehend offers several features:

- [Sentiment Analysis](#) - Determine the overall sentiment of text (positive, negative, or neutral).
- [Entity Recognition](#) - Identify named entities in text (e.g., people, places, organizations).
- [Keyword Extraction](#) - Extract important keywords from text.
- [Topic Modeling](#) - Automatically identify topics and themes in large collections of documents.
- [Summarization](#) - Generate concise summaries of text.

- Create a text file named  **PositiveSentiment.txt**, add the following text, and then save it in the **C:\Lesson01** folder:

 I absolutely love this product! It works perfectly and exceeded my expectations. I would definitely buy it again. You **MUST** try this.

Expand this hint for guidance on creating a text file.

- In the C:\Lesson01 folder, right-click, select **New**, and then select **Text Document**.
- Name the file  **PositiveSentiment.txt**, add the following text to the **PositiveSentiment.txt** file, and then **Save** it:

 I absolutely love this product! It works perfectly and exceeded my expectations. You **MUST** try this.

- Use AWS CLI for Sentiment Analysis.

💡 Expand this hint for guidance on using AWS CLI for Sentiment Analysis. ^

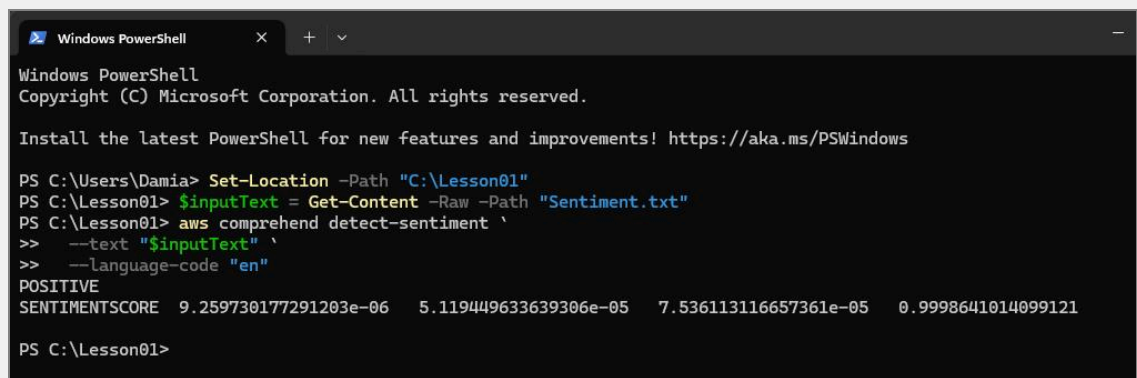
- If necessary, run Powershell as an Administrator, and then run the following command:

```
powershell
```

📄 **Set-Location -Path "C:\Lesson01"**

```
powershell
```

📄 **\$inputText = Get-Content -Raw -PositiveSentiment.txt**
aws comprehend detect-sentiment --text "\$(Get-Content -Raw -Path 'Positi



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Damia> Set-Location -Path "C:\Lesson01"
PS C:\Lesson01> $inputText = Get-Content -Raw -Path "Sentiment.txt"
PS C:\Lesson01> aws comprehend detect-sentiment `
>>   --text "$inputText" `
>>   --language-code "en"
POSITIVE
SENTENCESCORE 9.259730177291203e-06 5.119449633639306e-05 7.536113116657361e-05 0.9998641014099121
PS C:\Lesson01>
```

💡 Want to learn more? Review the documentation on [using AWS CLI for Sentiment Analysis](#).

- Record the first of the four Sentiment scores in the following text box.

Positive Sentiment Score

💡 Expand this hint for guidance on locating the Sentiment score. ^

- In the Windows Powershell console, locate the Positive Sentiment Score output, copy the first of the four scores, and then paste it into the Positive Sentiment Score box above.

💡 Want to learn more? Review the documentation on [locating the Sentiment score](#).

📄 Sentiment: **POSITIVE** ^

The analysis indicates that the overall sentiment of the text is positive. The SENTIMENTSCORE values represent the confidence scores for each of the sentiment categories (Positive, Negative, Neutral, Mixed), provided as floating-point numbers in scientific notation.

Here are the confidence scores broken down:

Positive: 9.259730177291203e-06 This score is a very small number (approximately 0.00000926), indicating a very low confidence that the sentiment is positive.

Negative: 5.119449633639306e-05 This score is also quite small (approximately 0.00005119), indicating a very low confidence that the sentiment is negative.

Neutral: 7.536113116657361e-05 This score is slightly higher but still small (approximately 0.00007536), indicating a low confidence that the sentiment is neutral.

Mixed: 0.9998641014099121 This score is very high (approximately 0.99986), indicating a very high confidence that the sentiment is mixed.

Overall, the analysis suggests that the sentiment of the text is predominantly positive, given the high confidence score for the mixed category.

- In the C:\Lesson01 folder, create a new python script named `NegativeSentiment.py` by using the following script, and then enter your **Access key** and **Secret access key**:

python



```
import boto3

# Initialize a session using Amazon Comprehend
session = boto3.Session(
    aws_access_key_id='YOUR_ACCESS_KEY_ID',
    aws_secret_access_key='YOUR_SECRET_ACCESS_KEY',
    region_name='us-east-1' # Specify the region
)

# Create a Comprehend client
comprehend = session.client(
    service_name='comprehend',
    endpoint_url='https://comprehend.us-east-1.amazonaws.com'
)

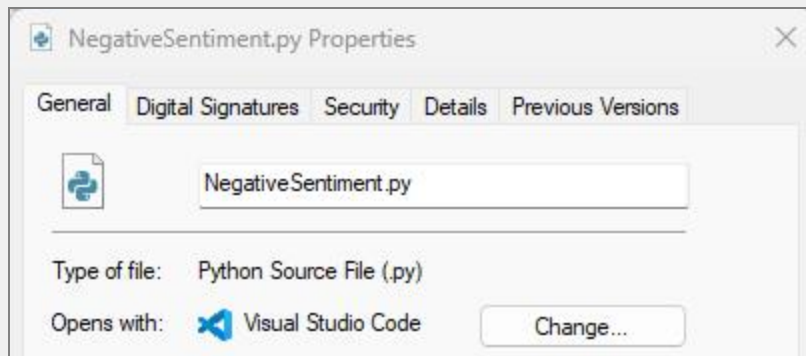
# Sample text for sentiment analysis
text = "I love using Amazon Comprehend for sentiment analysis!"

# Detect sentiment
response = comprehend.detect_sentiment(
    Text=text,
    LanguageCode='en'
)

# Print the sentiment
print(response)
```

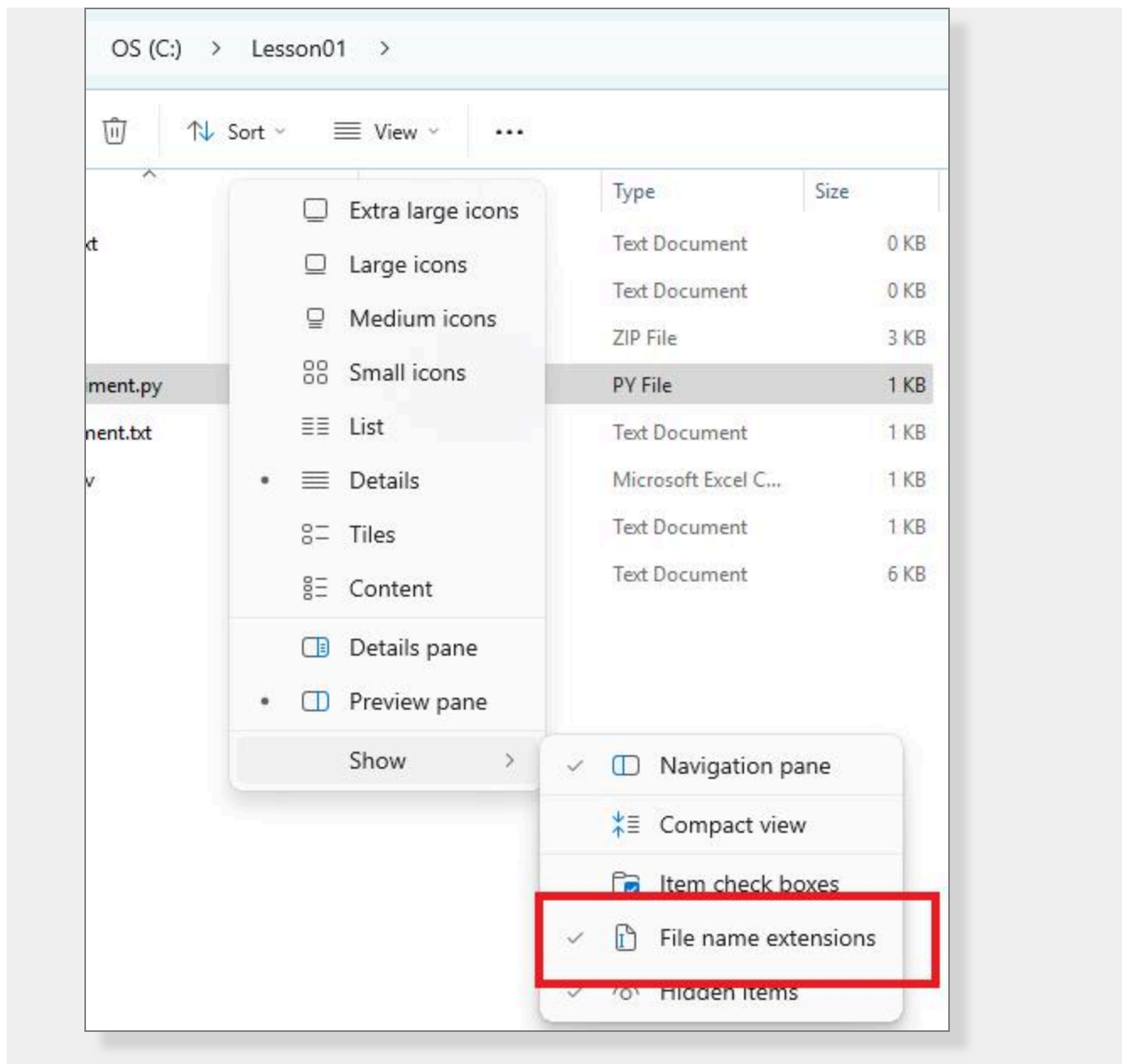

💡 Expand this hint for guidance on creating a python script. ^

- In the C:\Lesson01 folder, right-click, select **New**, and then select **Text Document**.
- Name the file `NegativeSentiment.py.txt`, add the above supplied script to the **NegativeSentiment.txt** file, and then **Save** it:
- Replace the `aws_access_key_id='YOUR_ACCESS_KEY_ID'` with your downloaded Access Key ID.
- Replace the `aws_secret_access_key='YOUR_SECRET_ACCESS_KEY'` with your downloaded Secret Access Key, and then save the **Sentiments.py** file.
- Right-click the file, select Properties, and then remove the **.txt** extension, so that the file name only has the **.py** file extension.



💡 Want to learn more? Review the documentation on [creating a python script](#).

📄 If you are unable to see the file extension which follows the file name, on the Menu bar, select **View > Show > File name extensions**. ^



- If necessary, install Python SDK, and then add it to your PATH.

 Expand this hint for guidance on installing Python SDK.

- In Windows Powershell, as an Administrator, enter the following to check the Python version:

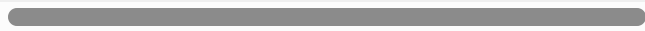
Powershell

 `python --version`

- If Python is not installed, in Windows Powershell, enter the following to download the Python installer:

Powershell

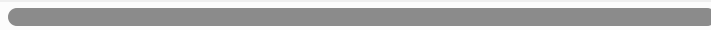
 `Invoke-WebRequest -Uri "https://www.python.org/ftp/python/3.9.7/python-3`

◀  ▶

- Run the Python installer by using the following command:

Powershell

 `Start-Process -FilePath "python-installer.exe" -ArgumentList "/quiet Ins`

◀  ▶

- On the do you want this app to make changes, select **Yes**.
- Add Python to your PATH ,replacing with the actual path to your Python installation, and then run the following command:

Powershell

 `$env:Path += ";<YourPythonPath>"`

- Verify the installation by using the following command:

Powershell

 `python --version` Want to learn more? Review the documentation on [installing Python SDK](#). If you are searching for the PATH to your Python directory, it might be something like

C:\Users\YourUsername\AppData\Local\Programs\Python\Python39.

- Use Python SDK for Sentiment Analysis.

 Expand this hint for guidance on using Python SDK for Sentiment Analysis. 

- In the Powershell terminal, run the following command to install the **boto3** library:

bash

 `pip install boto3`

- Run the following script:

bash

 `python negativesentiment.py`

- Your output should resemble the following:

```
PS C:\Lesson01> python NegativeSentiment.py
{'Sentiment': 'NEGATIVE', 'SentimentScore': {'Positive': 7.055841706460342e-05, 'Negative': 0.9998769760131836, 'Neutral': 3.666842894745059e-05, 'Mixed': 1.5792131307534873e-05}, 'ResponseMetadata': {'RequestId': 'acda4d75-dca2-467b-87e8-f8a167e7149c', 'HTTPStatusCode': 200, 'HTTPHeaders': {'x-amzn-requestid': 'acda4d75-dca2-467b-87e8-f8a167e7149c', 'content-type': 'application/x-amz-json-1.1', 'content-length': '166', 'date': 'Mon, 10 Feb 2025 20:05:49 GMT'}, 'RetryAttempts': 0}}
PS C:\Lesson01>
```

 Want to learn more? Review the documentation on [using Python SDK for Sentiment Analysis](#).

 Your screen should look like the following: 

```
PS C:\lesson01> python sentiment.py
{'Sentiment': 'NEGATIVE', 'SentimentScore': {'Positive': 7.055841706460342e-05, 'Negative': 0.9998769760131836, 'Neutral': 3.666842894745059e-05, 'Mixed': 1.5792131307534873e-05}, 'ResponseMetadata': {'RequestId': '24f9a081-e809-40ba-88ec-29ebe27bce28', 'HTTPStatusCode': 200, 'HTTPHeaders': {'x-amzn-requestid': '24f9a081-e809-40ba-88ec-29ebe27bce28', 'content-type': 'application/x-amz-json-1.1', 'content-length': '166', 'date': 'Fri, 07 Feb 2025 03:15:29 GMT'}, 'RetryAttempts': 0}}
PS C:\lesson01>
```

- Record the Negative Sentiment score in the following text box.

Negative Sentiment Score



Expand this hint for guidance on locating the Sentiment score.



- In the Windows Powershell console, locate the Negative Sentiment Score output, copy the Negative Sentiment value, nested between the Postive and Neutral scores, and then paste it into the Positive Sentiment Score box above.



Here's a breakdown of the response from AWS Comprehend:

Sentiment: 'NEGATIVE'

This indicates that the overall sentiment of the text analyzed is negative.

[more...](#)

Check your work

Verify

Perform Entity Recognition to identify named entities in text

Hints Enabled

No Yes

- Create a text file named **Entities.txt**, add the following text, and then save it in the **C:\Lesson01** folder:

Entities.txt Hexelo Inc. announced a new AI elearning model at the annual Artificial Intelligence convention at World Theater in Chicago, Illinois last week.

 Expand this hint for guidance on creating a text file. 

- In the C:\Lesson01 folder, right-click, select **New**, and then select **Text Document**.
- Name the file **Entities.txt**, add the following text to the **Entities.txt** file, and then **Save** it:

Entities.txt Hexelo Inc. announced a new AI elearning model at the annual Artificial Intelligence convention at World Theater in Chicago, Illinois last week.

 Want to learn more? Review the documentation on [creating a text file](#).

- Use AWS CLI for Entity Recognition by running the following script:

bash

```
 aws comprehend detect-entities `
--text "C:\Lesson01\entities.txt" `
--language-code "en"
```

💡 Expand this hint for guidance on using AWS CLI for Entity Recognition. ^

Run the following script:

```
bash
aws comprehend detect-entities `
  --text "C:\Lesson01\entities.txt" `
  --language-code "en"
```

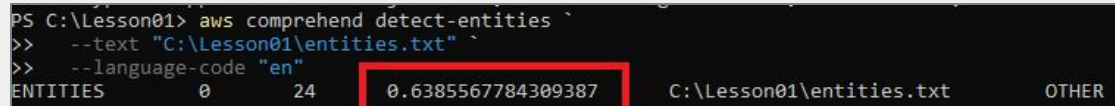
💡 Want to learn more? Review the documentation on [using AWS CLI for Entity Recognition](#).

- Record the confidence score in the following text box.

Confidence Score

💡 Expand this hint for guidance on locating the Confidence score. ^

- In the Windows Powershell console, locate the Confidence Score output, copy and then paste it into the Confidence Score box above.



```
PS C:\Lesson01> aws comprehend detect-entities `
>> --text "C:\Lesson01\entities.txt" `
>> --language-code "en"
ENTITIES      0      24      0.6385567784309387      C:\Lesson01\entities.txt      OTHER
```

- In the C:\Lesson01 folder, create a new python script named `Entities.py` by using the following script:

python

```
import boto3

comprehend = boto3.client('comprehend', region_name='us-east-1')

text = "Hexelo Inc. announced a new AI eLearning model at the annual Artif
response = comprehend.detect_entities(
    Text=text,
    LanguageCode='en'
)

print("Detected Entities:")
```

```
for entity in response['Entities']:
    print(f"- {entity['Text']} ({entity['Type']})")
```

💡 Expand this hint for guidance on creating a python script. ^

- In the C:\Lesson01 folder, right-click, select **New**, and then select **Text Document**.
- Name the file `Entities.py.txt`, add the above supplied script to the **Entities.txt** file, and then **Save** it:
- Right-click the file, select Properties, and then remove the **.txt** extension, so that the file name only has the **.py** file extension.

- Use Python SDK for Entity Recognition.

💡 Expand this hint for guidance on using Python SDK for Entity Recognition. ^

- Run the following script:

bash

 python entities.py

- Record the Event that is listed multiple times in the following text box.

Event

💡 Expand this hint for guidance on locating the Confidence score. ^

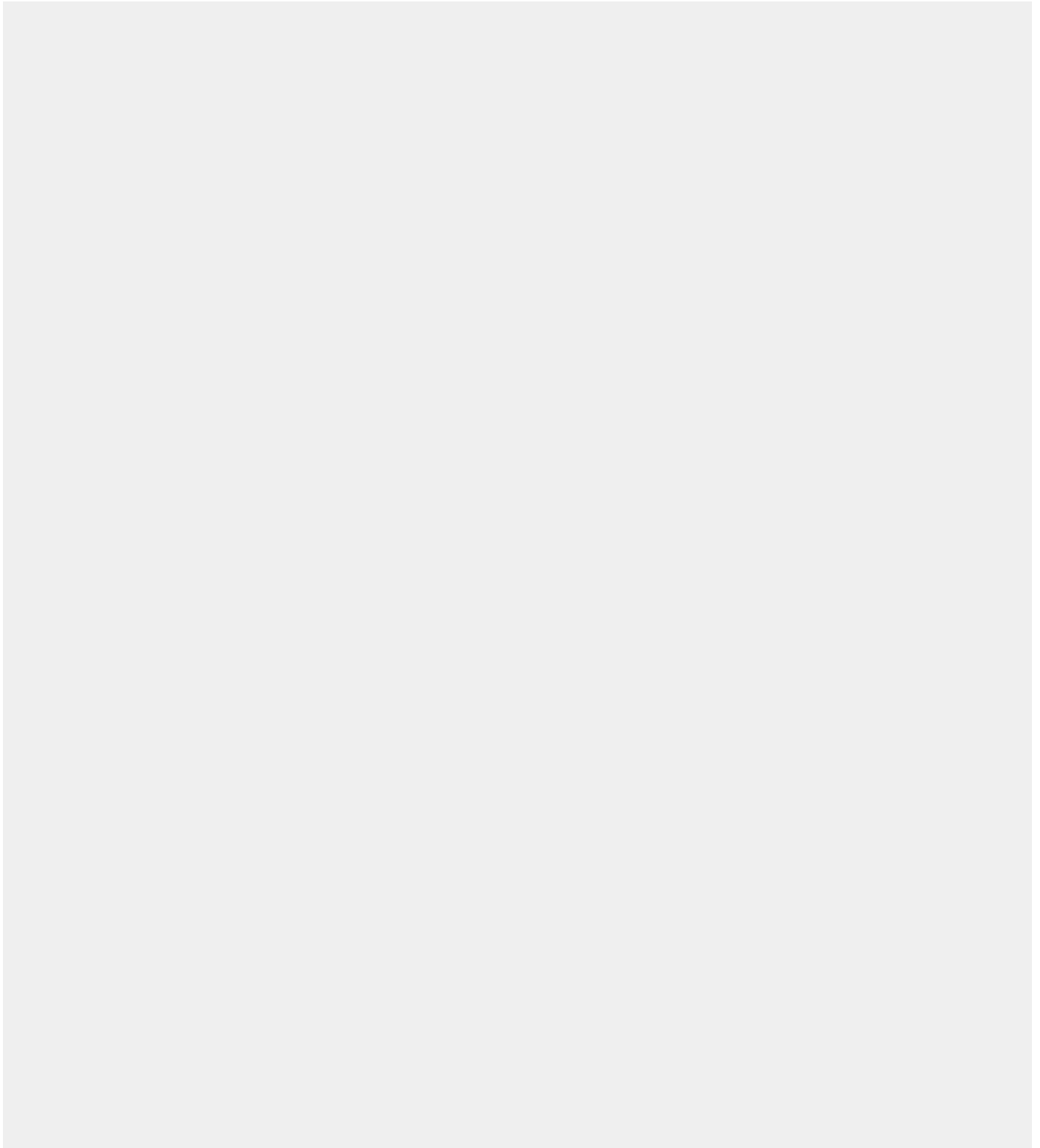
- In the Windows Powershell console, locate the Detected Entities output, and then copy and paste the repeated entity into the Entity box above.

```
PS C:\Lesson01> python entities.py
Detected Entities:
- Hexelo Inc. (ORGANIZATION)
- Artificial Intelligence convention (EVENT)
- World Theater (LOCATION)
- Chicago, Illinois (LOCATION)
- last week (DATE)
PS C:\Lesson01>
```


Check your work



Verify



Extract keywords from text by using Amazon Comprehend

Hints Enabled

No Yes

- Create a text file named `keywords.txt` by using the following text:

The impact of artificial intelligence on IT professionals has been significant due to increased productivity.

 Expand this hint for guidance on creating a text file. 

- Create a text file named `keywords.txt` by using the following text:

The impact of artificial intelligence on IT professionals has been significant due to increased productivity.

- Open an **Administer: Command Prompt** and use AWS CLI for Keyword Extraction by running the following script..

```
bash
```

```
 aws comprehend detect-keyphrases --text file://C:/Lesson01/keywords.txt --1
```

 Expand this hint for guidance on using AWS CLI for Keyword Extraction.

- Open a new Command Prompt, run as an Administrator, change to the keywords.txt directory, and then run the following command:

bash

 `aws comprehend detect-keyphrases --text file:///C:/Lesson01/keywords.txt`

- Your output should resemble the following:

```
c:\Lesson01>aws comprehend detect-key-phrases --text file:///C:/Lesson01/keywords.txt --language-code en
KEYPHRASES      0      10      0.9999864101409912      The impact
KEYPHRASES     14      37      0.9999714493751526      artificial intelligence
KEYPHRASES     41      57      0.9995271563529968      IT professionals
KEYPHRASES     86     108      0.9962848424911499      increased productivity
```

 Want to learn more? Review the documentation on [using AWS CLI for Keyword Extraction](#).

- Count, and then record the number of KEYPHRASES in the following text box:

KEYPHRASES

 Expand this hint for guidance on locating the KEYPHRASES.

- In the Command Prompt, locate the detect-key-phrases output, and then count and paste the number of keyphrases into the KEYPHRASES box above.

- In the C:/Lesson01 folder, create a new python script named `keywords.py` by using the following script:

python



```
import boto3

comprehend = boto3.client('comprehend', region_name='us-east-1')

text = "The impact of artificial intelligence on IT professionals has been
response = comprehend.detect_key_phrases(Text=text, LanguageCode='en')

print("Detected Keywords:")
```

```
for phrase in response['KeyPhrases']:
    print(f"- {phrase['Text']} (Score: {phrase['Score']:.2f})")
```

 Expand this hint for guidance on creating a python script. 

- In the C:\Lesson01 folder, right-click, select **New**, and then select **Text Document**.
- Name the file  **keywords.py.txt**, add the above supplied script to the **keywords.txt** file, and then **Save** it:
- Right-click the file, select Properties, and then remove the **.txt** extension, so that the file name only has the **.py** file extension.

- Use Python SDK for Keyword Extraction.

 Expand this hint for guidance on using Python SDK for Keyword Extraction. 

- Run the following script to use Python SDK for Keyword Extraction:

bash

 python keywords.py

 Want to learn more? Review the documentation on [using Python SDK for Keyword Extraction](#).

- Count, and then record the number of KEYPHRASES in the following text box:

KEYWORDS

 Expand this hint for guidance on locating the Detected Keywords. 

- In the Windows Powershell console, locate the impact score, and then enter it, including all decimals, into the KEYWORDS box above.

 Want to learn more? Review the documentation on [locating the Detected Keyword](#).

Check your work

Verify

Perform Topic Modeling

Hints Enabled

No Yes

 To perform **Topic Modeling**, you need a collection of documents. For simplicity, we'll use two sample documents.

- Create a new folder named  **AWS Documents** in the **C:\Lesson01** folder, add 2 text files named  **doc1** and  **doc2**, and then add the following to each document:
 - **Doc1:**  This is a document about artificial intelligence.
 - **Doc2:**  Another document discussing artificial intelligence.

 Expand this hint for guidance on creating a folder. 

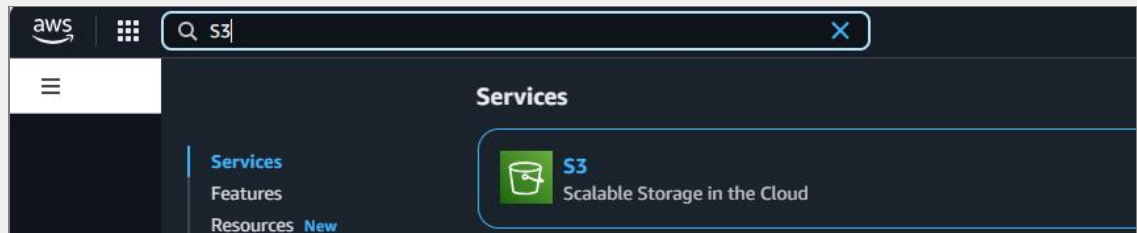
- In the **C:\Lesson01** folder, right-click, select **New**, and then select **Folder**.
- Name the folder named  **AWS Documents**, and then select **Enter**.
- In the **C:\Lesson01** folder, right-click, select **New**, select **Text Document**, and then name the file  **doc1.txt**.
- Add the following text to **doc1**, and then save the file:
 This is a document about artificial intelligence.
- Repeat the previous step to create  **doc2.txt**.
- Add the following text to **doc2**, and then save the file:
 Another document discussing artificial intelligence.

 You'll need to upload your documents to an S3 bucket first.

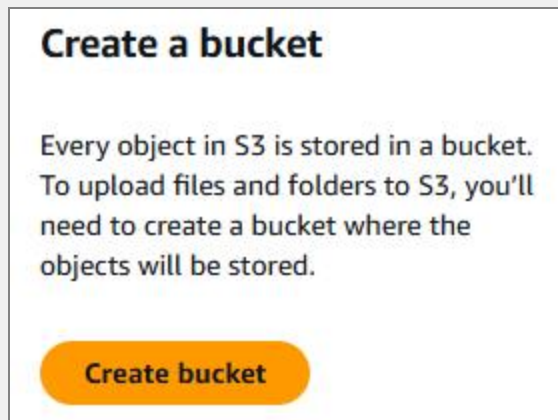
- Create an Amazon S3 bucket named  **tm.input**.

💡 Expand this hint for guidance on creating an S3 bucket.

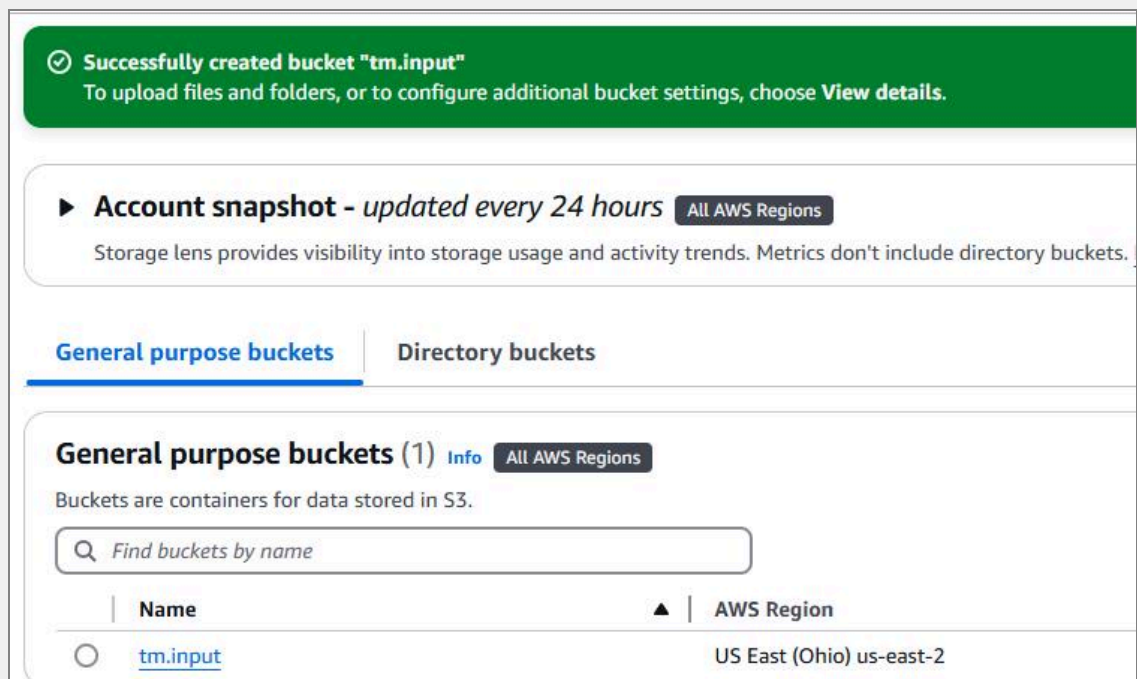
- In the Services menu, search for and select **T** S3.



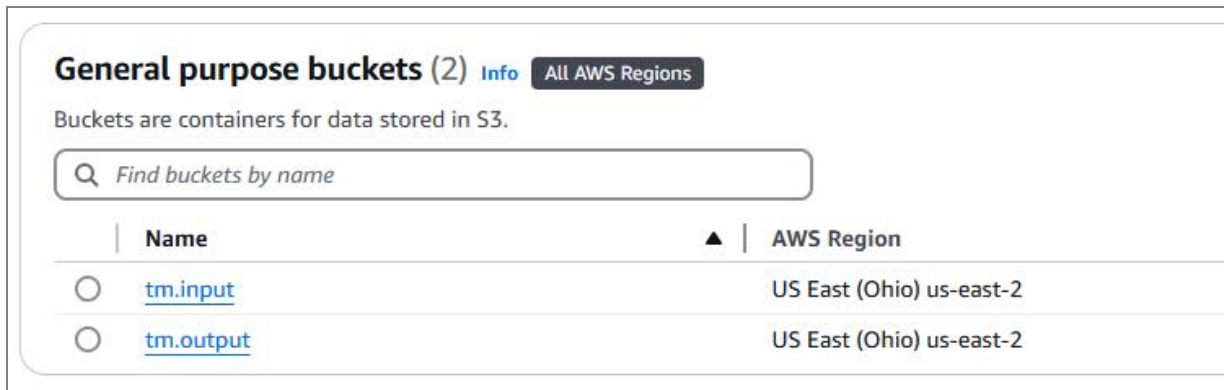
- On the Amazon S3 home page, select **Create bucket**.



- On the Create bucket page, in General configuration, in Bucket name, enter **T** `tm.input`, and then select **Create bucket**.



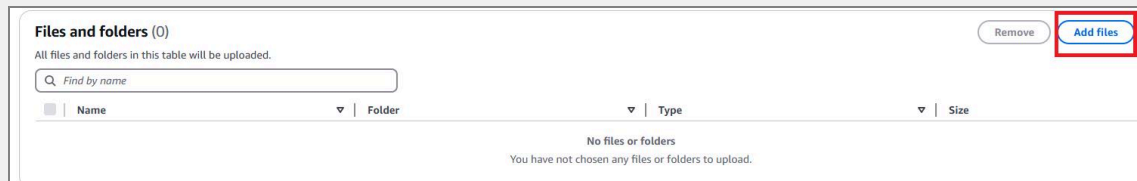
- Create a second Amazon S3 bucket named **tm.output**.



- Upload **doc1** and **doc2** to the **tm.input** S3 bucket.

💡 Expand this hint for guidance on uploading a file to an S3 bucket.

- On the Buckets page, in General purpose buckets, select the **tm.input** bucket (the link, not the radio button), and then select **Upload**.
- On the Upload page, in Files and folders, select **Add files**.



Files and folders (0)

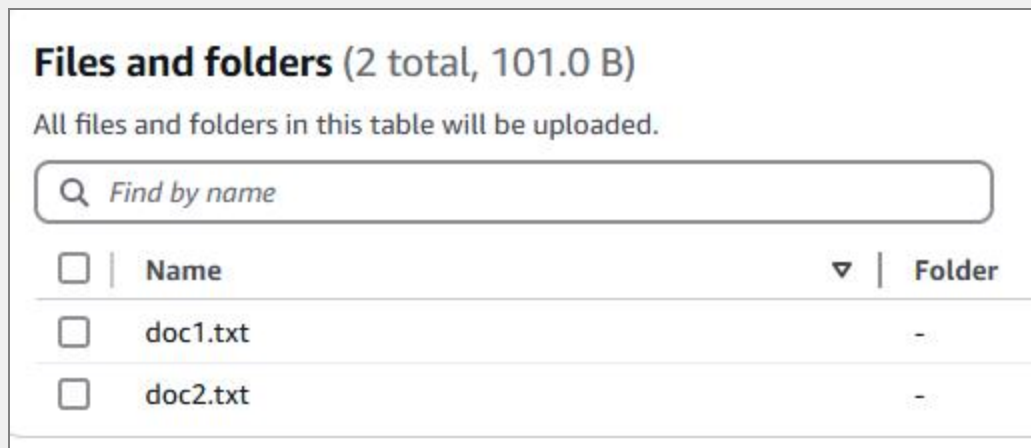
All files and folders in this table will be uploaded.

Find by name

☐	Name	Folder	Type	Size
No files or folders				
You have not chosen any files or folders to upload.				

Remove Add Files

- Navigate to your T C:/Lesson01/AWS Documents folder, select **doc1** and **doc2**, and then select **Open**.



Files and folders (2 total, 101.0 B)

All files and folders in this table will be uploaded.

Find by name

☐	Name	Folder
☐	doc1.txt	-
☐	doc2.txt	-

- On the Upload page, select **Upload**.

Your screen should resemble the following:

✓ Upload succeeded
For more information, see the [Files and folders](#) table.

Upload: status

ⓘ After you navigate away from this page, the following information is no longer available.

Summary

Destination
s3://tm.input

Succeeded
✓ 2 files, 101.0 B (100.00%)

Files and folders

Configuration

Files and folders (2 total, 101.0 B)

🔍 Find by name

Name	Folder	Type
doc1.txt	-	text/plain
doc2.txt	-	text/plain

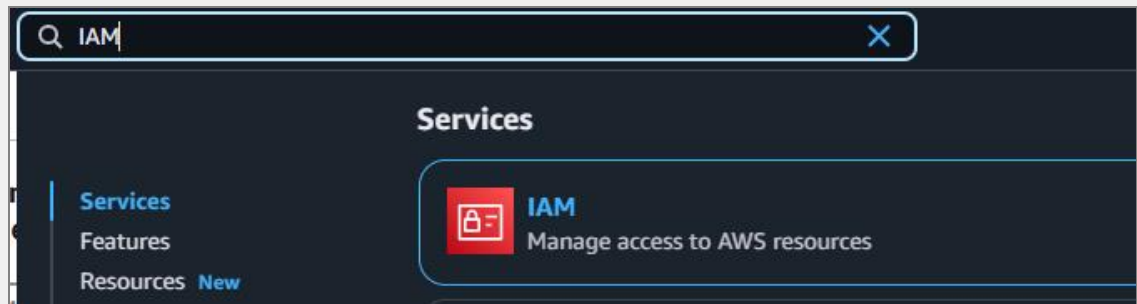
- Create an IAM Role for Amazon Comprehend named **ComprehendS3AccessRole**, and then Attach AmazonS3FullAccess and ComprehendFullAccess policies

<https://labondemand.com/LabProfile/Instructions/181230?instructionsSetId=284852>

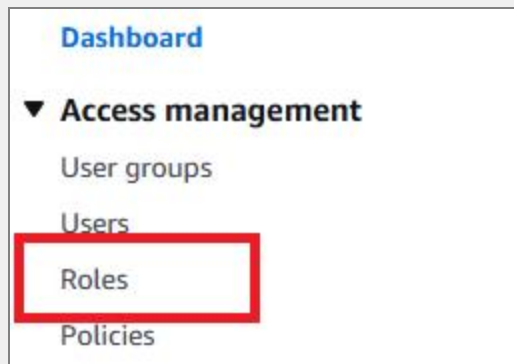
34/43

💡 Expand this hint for guidance on creating an IAM Role for Amazon Comprehend. ^

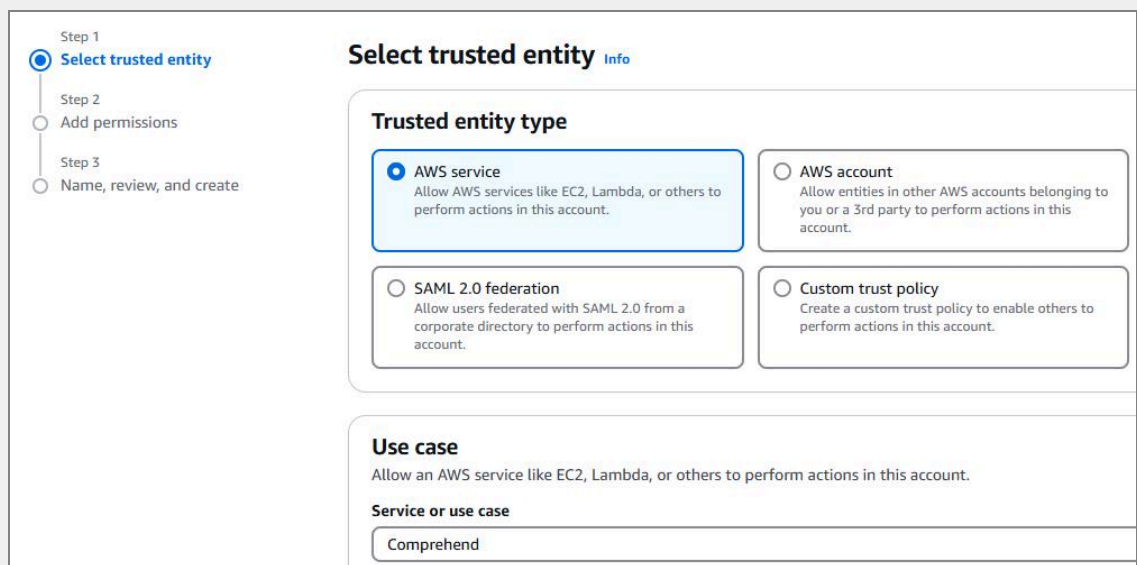
- In the Services menu, search for and select **IAM**.



- On the IAM Dashboard, in Access management, select **Roles**.



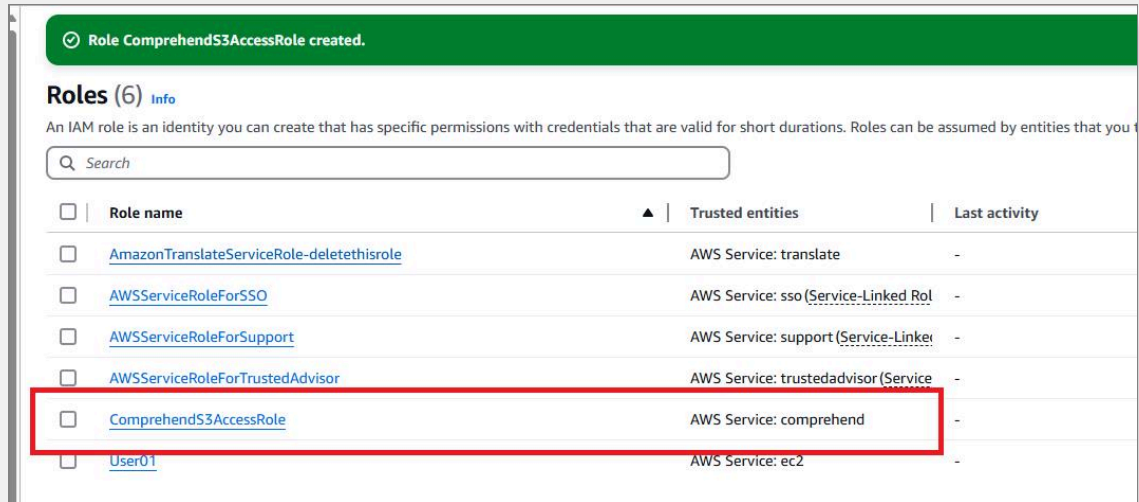
- On the Roles page, select **Create role**
- On the Create role | Select trusted entity page, in Trusted entity type, select **AWS service**, and then in Use case, search for and select **Comprehend**, and then select **Next**.



- On the Add permissions page, in Set permissions boundary-optional, select **Use a permissions boundary to control the maximum role permissions**, and then in

Permissions policies, search for and select **AmazonS3FullAccess** and **ComprehendFullAccess**.

- Select **Next**, and then on the Role details page, in Role name, enter **ComprehendS3AccessRole**.
- Select **Create role**.

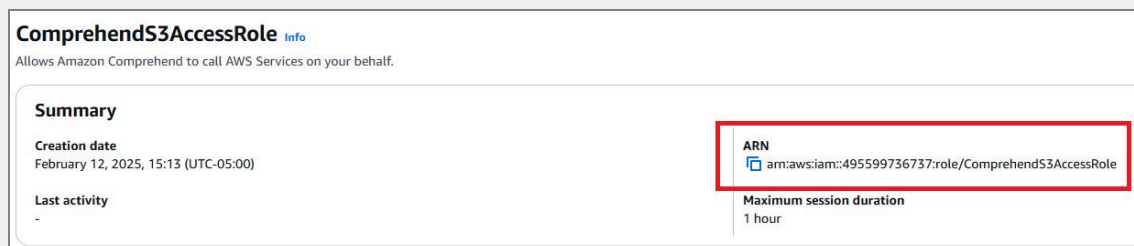


- Record the ARN of the role in the following text box.

ARN role

Expand this hint for guidance on locating the ARN role.

- On the **Identity and Access management (IAM) | Roles** page, in Roles, select **ComprehendS3AccessRole**.
- On the ComprehendS3AccessRole page, in Summary, locate the ARN, and then select the copy-to-clipboard icon.



- Paste the ARN into the textbox above.

- Use AWS CLI for Topic Modeling by running the following script, by using the **ComprehendS3AccessRole** ARN:

bash

```
aws comprehend start-topics-detection-job \
  --data-access-role-arn <YOUR-ARN-GOES-HERE> \
  --input-data-config S3Uri=s3://tm.input/ \
  --output-data-config S3Uri=s3://tm.output/ \
  --number-of-topics 10 \
  --region us-east-1
```

 Expand this hint for guidance on using AWS CLI for Topic Modeling. 

- Open a command prompt window, and then run the following command to start a topic modeling job:

bash

```
aws comprehend start-topics-detection-job \
  --data-access-role-arn <YOUR-ARN-GOES-HERE> \
  --input-data-config S3Uri=s3://tm.input/ \
  --output-data-config S3Uri=s3://tm.output/ \
  --number-of-topics 10 \
  --region us-east-1
```

 Want to learn more? Review the documentation on [using AWS CLI for Topic Modeling](#).

- Record the single word response, in all caps, in the following text box.

Status

 Expand this hint for guidance on locating the Confidence score. 

- In the Command prompt console, locate the very last word, displayed in all capital letters, and then copy and paste it into the Status box above.

```
C:\Lesson01>aws comprehend start-topics-detection-job --data-access-role-arn arn:aws:iam::495599736737:role/ComprehendS3AccessRole --input-data-config S3Uri=s3://tm.input/ --output-data-config S3Uri=s3://tm.output/ --number-of-topics 10 --region us-east-1
arn:aws:comprehend:us-east-1:495599736737:topics-detection-job/b57b0d5cc4b706eda55f065359fa3767 b57b0d5cc4b706eda55f065359fa3767 SUBMITTED
```

- Check Job Status.



Expand this hint for guidance on checking the job status.



- Run:

bash



```
aws comprehend describe-topic-detection-job \  
--job-id your-job-id
```



Want to learn more? Review the documentation on [checking the job status](#).



Once the job is complete, you can download the results from the output S3 bucket.

Check your work

Activity Group requirement5 could not be found.

Use the Summarization feature to generate a concise summary of text

Hints Enabled

No Yes

- Create a text file named `summary.txt` by using the following text:

The rapid development of artificial intelligence has revolutionized many industries. From healthcare to finance, AI is being used to improve decision-making and automate processes. Companies like Google, Microsoft, and IBM are leading the way in AI research and development.

 Expand this hint for guidance on creating a text file. 

- Create a text file named `summary.txt` by using the following text:

The rapid development of artificial intelligence has revolutionized many industries. From healthcare to finance, AI is being used to improve decision-making and automate processes. Companies like Google, Microsoft, and IBM are leading the way in AI research and development.

 Want to learn more? Review the documentation on [creating a text file](#).

- Use AWS CLI for Summarization.

💡 Expand this hint for guidance on using AWS CLI for Summarization. ^

- Run:

bash

```
aws comprehend summarize-text \  
  --text file://$(pwd)/summary.txt \  
  --number-of-sentences 1
```

Output Example:

json

```
{  
  "Summary": [  
    {  
      "Text": "The rapid development of artificial intelligence has    }  
  ]  
}
```

💡 Want to learn more? Review the documentation on [using AWS CLI for Summarization](#).

- Use Python SDK for Summarization.

💡 Expand this hint for guidance on using AWS CLI for Summarization. ^

Create a script ( summarize.py):

```
python
import boto3

comprehend = boto3.client('comprehend', region_name='your-region')

text = "The rapid development of artificial intelligence has revolutionized
automate processes. Companies like Google, Microsoft, and IBM are leading

response = comprehend.summarize_text(
    Text=text,
    MaxSentences=1
)

print("Summary:", response['Summary'][0]['Text'])
```

- Run the script:

```
bash
python summarize.py
```

💡 Want to learn more? Review the documentation on [using AWS CLI for Summarization](#).

⚠ Always ensure you comply with AWS pricing policies when using these services, as they may incur charges based on usage.

Check your work

Activity Group requirement6 could not be found.

Summary

Congratulations, you have completed the **Perform Natural Language Processing by Using Amazon Comprehend** Challenge Lab.

You have accomplished the following:

- Set Up Your AWS Account and IAM Role.
- Performed Sentiment Analysis.
- Tried Entity Recognition to identify named entities in text.
- Extracted keywords from text using Amazon Comprehend.
- Performed Topic Modeling.
- Used the Summarization feature to generate a concise summary of text.

Ending your lab

To ensure your lab is recorded as complete, select **Submit** or **End** below. Exiting [X] the lab will result in an incomplete status.

 Once you select **Submit** or **End**, you will not be able to return to this Challenge Lab.

Your feedback is important!

As you end your Challenge Lab, please take a few minutes to complete the short survey that will appear in the next window. Alternatively, you may provide your feedback directly to [Challenge Labs feedback](#).

Looking for your next Challenge Lab?

Below are recommended Challenge Labs to try next.

